

CCNA 200-301 - Chapter 9: Subnetting Part 1

Subnetting is the process of dividing one large network into smaller subnetworks.

Benefits

- Performance: Reduce Broadcast traffic.
- Security: Isolate departments.
- Organization: Easier management.

Scenario

HR = 192.168.1.0/24

Sales = 192.168.2.0/24

Accounting = 192.168.3.0/24

This is NOT true subnetting because only the network addresses changed while the subnet mask stayed the same.

Real Subnetting

Subnetting changes the subnet mask (e.g. /24 to /26).

Inter-Subnet Communication

Different subnets communicate using a Router or Layer 3 Switch.

ACL

ACL rules control which departments can communicate.